What Prices Cannot Tell You: Identifying Information Transmission in Live Markets

A companion study asked whether prices contain public information. This paper asks whether prices reveal how that information entered the market.

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> **Draft status.** This manuscript is complete through the Methods section. The Results and Discussion sections are deliberately unwritten: they are gated on four objective measurement conditions stated in Section 6.6, and the analysis plan in Section 6.5 is pre-registered so that the design cannot be shaped by the evidence. Nothing in the sections below reports an empirical finding from the dataset described in Section 5.

Abstract

A companion study found that a live baseball wagering market encompasses every public-information candidate tested against it: no variable in a ten-hypothesis ladder improved on the market's own forecast of remaining runs. That null is a statement about outcomes, and it invites a question about mechanism. The companion study asked whether prices *contain* public information; this paper asks whether prices *reveal how that information entered the market*. Information in prices and formation of prices are distinct questions, and the second does not follow from the first by adding data.

We take up the transmission question directly, and find that it is largely an identification problem. The delay between a game event and an observed price change decomposes into a bookmaker's pricing latency, its feed's transport latency, and the observation latency imposed by our own sampling. That decomposition is arithmetic. The substantive difficulty is that only their sum is observed, and this paper is about what can and cannot be learned under that constraint. We show that three materially different markets, one in which a bookmaker genuinely prices faster, one in which pricing speeds are identical and only publishing infrastructure differs, and one in which both differ and partially offset, generate numerically identical timestamp data. No statistic computed from timestamps alone separates them.

Working from a two-book, thirty-one-second live quote panel matched to game-state events, we define an event-anchored response latency and state precisely the conditions under which its cross-book contrast is a statement about pricing rather than about plumbing. We show that several intuitive estimators of information leadership are mechanically confounded by update frequency and by the analyst's choice of which posted line to treat as the main line, and we characterize the resolution floor below which this class of instrument is silent. The design admits three outcomes, all of which we regard as publishable: identification of a pricing contrast, identification of bounds only, or a demonstration that the quantity is not identifiable without richer instrumentation. The price discovery literature

answers our question directly where venue timestamps and order identifiers are available, because there the publication stage is pinned by the data rather than estimated; the contribution here is to characterize what remains learnable when it is not. The third is not a failure of the study. Much of the microstructure literature studies environments in which richer event and timestamp information is available than sportsbook endpoints provide, so the decomposition we confront can often be assumed away there rather than established. Showing precisely where it cannot be assumed away is itself a contribution.

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1. Introduction

Every pitch thrown in a Major League Baseball game can trigger dozens of price updates across sportsbooks before the next pitch is delivered. These are high-frequency markets in the ordinary sense, with one property that makes them an unusually clean laboratory for studying price formation: every contract settles within hours against an objective terminal payoff, so the quantity being forecast is eventually revealed and the forecast can be graded. Equity prices offer no such closure.

Market efficiency tests answer a question about outcomes. They ask whether some information, held by someone, would have improved on the price. When the answer is no, as it frequently is, the finding is reported and the matter closed. But a null of that kind is a statement about a mechanism whose operation was never observed. Something incorporated the information. The efficiency test does not say what, or how quickly, or through which participant.

This paper begins where a companion study ended. That study treated a sharp live betting line as an incumbent forecast of the runs remaining in a baseball game and asked whether any of ten publicly observable candidates, pitcher fatigue and velocity decay among them, carried incremental predictive content beyond it. None did, and the one candidate that appeared alive proved to be a post-treatment selection artifact.

The two studies stand in a natural progression rather than a sequence of increments. **Paper 1 asked whether prices contain public information. This paper asks whether prices reveal how that information entered the market.** The first is a question about the content of a price; the second is a question about its formation, and the two are answered with different instruments and different assumptions. The transition can be stated more precisely still. **The companion study established that public baseball variables contain no incremental information beyond a sharp market observed at one-minute resolution; this paper asks whether that apparent efficiency is genuine, or whether it is partly an artifact of observing the market only after the information has already propagated through it.** The companion study's closing section anticipates the question, listing cross-book propagation, the information half-life of a shock, and the microstructure limits of one-minute single-book snapshots among the things its data could not settle.

The obstacle is not statistical power. It is that the process of interest is unobserved. A baseball market is punctuated by discrete, publicly dated events, and each is followed, sooner or later, by a revised price; but between the event and the revision we record sit stages that no outside observer sees.

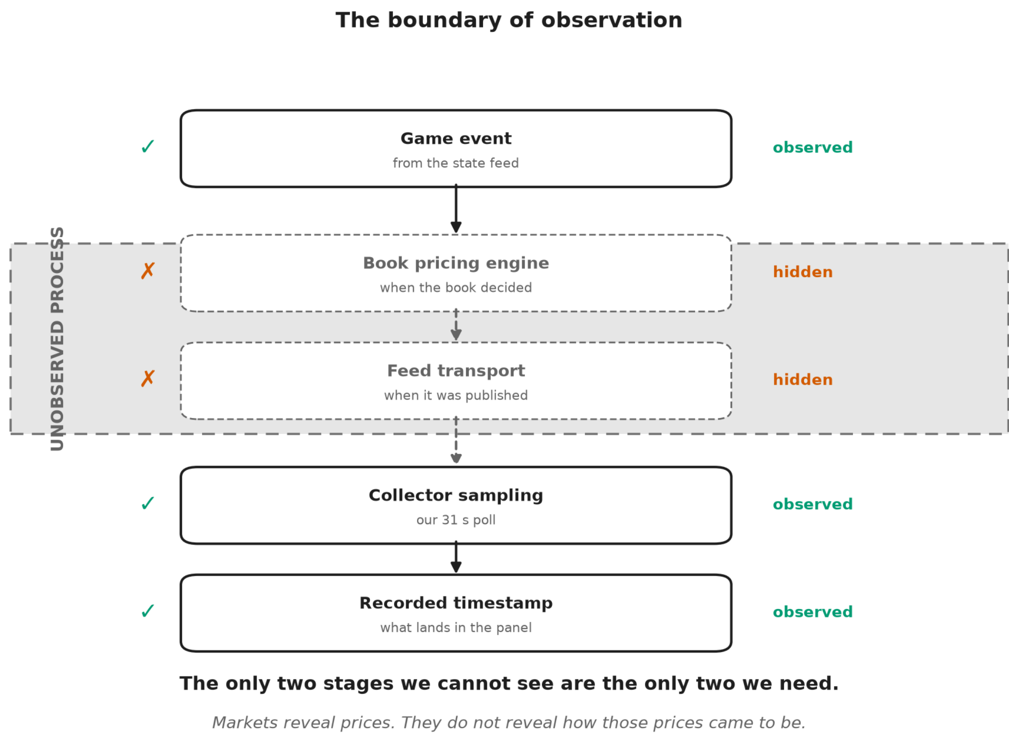

Figure 1. *We never observe the pricing process itself.* Of the five stages between a game event and a row in our dataset, the two in the shaded band are exactly the two the question is about: when the bookmaker decided to move, and when its feed made that decision visible. Everything we record sits below the boundary.

Box 1 makes the difficulty concrete before any notation is introduced.

> **Box 1. A thought experiment**
>
> Two sportsbooks are quoting the same game. A run scores. Book A updates its price 800 milliseconds later; Book B updates 1.6 seconds later. You observe both prices once every 30 seconds.
>
> Can you conclude that Book A processes information faster?
>
> **No.** You did not observe the pricing engines, the feed delays, or the moment the event actually occurred. You observed two timestamps, produced by a sampling process of your own construction, at a resolution nearly twenty times coarser than the difference you are trying to detect. Both books land in the same poll. The data are silent.
>
> This paper is about what it takes to make them speak, and about being candid when they cannot.

The problem has a familiar shape. Astronomers infer the mass of a planet they cannot see from the wobble of a star they can, and the inference is only as good as the model of what lies between the observer and the object; much of the work is characterizing the instrument rather than the sky. Our position is the same. **Markets reveal prices. They do not reveal how those prices came to be.**

Two prices are therefore the minimum. A single book's response time confounds how fast the bookmaker reprices with how fast its infrastructure publishes and how fast the analyst happens to be sampling. Comparing two books raises the possibility that the shared components difference out. Whether they actually do is an architectural question about publishing infrastructure, not a statistical one, and it is the question on which the paper's central estimand stands or falls. Section 4 shows that three materially different markets generate numerically identical timestamp data, so the possibility cannot be settled by any statistic computed from the timestamps themselves; that result, illustrated in Figure 4, is the paper's core.

We frame this accordingly as a paper about identification in high-frequency markets, not as a second efficiency paper and not as a paper about sportsbooks. Sports betting is the laboratory, chosen because its information arrivals are discrete and precisely dated and its contracts settle against ground truth. The identification problem it exposes is general. Three consequences follow.

First, the estimand must be defined before the data are examined, because the intuitive estimators are wrong in a specific and instructive way. The obvious approach, timing one book's price change against the other's, is mechanically biased toward whichever book updates more often. A book that re-prices every thirty seconds will reach any given price level before a book that re-prices every eight minutes, regardless of which one is better informed. That is not leadership; it is sampling.

Second, the identification problem must be confronted rather than assumed away. The delay we observe is a sum of three delays, and only one of them is about pricing. Separating them requires either an independent measurement of transport latency or an argument that transport latency is common across books. Neither is free, and we do not assume either.

Third, the instrument's limits must be stated in advance. A panel sampled every thirty-one seconds cannot resolve sub-second price formation. If the economically interesting action happens below the sampling floor, no amount of care in the analysis will recover it, and the honest report is that the question is out of reach for this apparatus.

The paper is organized around those three commitments. Section 3 develops the conceptual framework and decomposes the observed delay. Section 4, the heart of the paper, treats identification: what can and cannot be learned from a two-book panel, and under exactly which assumptions. Section 4 describes the data and explains why this instrument differs fundamentally from the one used in the companion study, a difference that precludes any direct comparison of results. Section 5 states the estimation procedure and the pre-registered analysis plan, and closes by listing the objective conditions that must be satisfied before results may be reported.

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2. Related literature and the identification gap

A referee's first question about a paper like this is why the problem has not already been solved. The answer is not that the literatures below overlooked it. It is that in the settings where price discovery has been studied most carefully, the instrumentation available to the researcher resolves the problem by construction, and the question therefore never has to be posed. We organize this section around what each literature identifies, what data it identifies it with, and what happens to the argument when those data are unavailable.

2.1 What prices contain

The efficiency literature asks whether prices already embody available information. Fama (1970) sets the frame; the event-study tradition sharpened it to the speed of adjustment, with Patell and Wolfson (1984) measuring intraday adjustment to earnings and dividend announcements and Busse and Green (2002) tracking incorporation of televised analyst reports in real time. In wagering markets the same question has an unusually clean form because contracts settle against ground truth: Sauer (1998) surveys the evidence, Woodland and Woodland (1994) document a reverse favorite-longshot bias in baseball too small to exploit, and Thaler and Ziemba (1988) place the setting in the anomalies tradition. Croxson and Reade (2014) show soccer exchange prices updating swiftly and essentially fully on goal arrivals. More recent work finds imperfections in the price process itself: Simon (2024, 2025) rejects weak-form efficiency in moneyline movement, and Angelini and De Angelis (2026) measure contemporaneous underreaction to benchmark probability changes with predictable subsequent drift.

These studies identify a property of the price. None of them requires knowing *when the bookmaker or the exchange internally decided to move*, because the object of inference is the price level and its relation to information, not the timing of the mechanism that produced it. Our companion study sits squarely here: it applies the forecast encompassing framework of Chong and Hendry (1986) to a live betting line treated as an incumbent forecast, and its null is a statement of the same type.

2.2 How information enters prices

The price discovery literature asks where a common efficient price is formed when the same asset trades in several venues. Hasbrouck's (1995) information shares and the Gonzalo and Granger (1995) permanent-transitory decomposition are the standard tools, and Putniņš (2013) offers a careful account of what those metrics do and do not measure. This literature is the closest ancestor of the present paper: it asks precisely our question, which venue moves first and why.

Its instrumentation is what makes it work. Information shares are estimated on venue-level quote and trade series carrying exchange-side timestamps, applied to a common clock, at a granularity fine relative to the leads being estimated. The publication stage is not a nuisance term to be estimated because it is pinned directly by the venue's own timestamp. Where that is unavailable, the metric does not become noisier; it becomes a different quantity.

2.3 Timing across markets

A related tradition estimates leads and lags directly. Hasbrouck (1991) recovers the information content of trades from a bivariate vector autoregression; Chan (1992) analyzes the lead-lag relation between index futures and the cash market. Later work pushes to the latency frontier: Hasbrouck and Saar (2013) identify low-latency strategic activity using millisecond-stamped order-level messages, and Budish, Cramton, and Shim (2015) analyze the speed race itself using synchronized data on correlated instruments.

Two features of these designs deserve emphasis, because their absence defines our problem. First, order-level identifiers let the researcher follow an individual message rather than infer activity from a price series. Second, the timestamps are applied at the venue, so transport to the researcher does not enter the estimand. Ding, Hanna, and Hendershott (2014) make the second point especially clearly: they compare the consolidated feed with direct exchange feeds and quantify the dissemination latency between them, which is possible precisely because both feeds are observable. Their result is the strongest evidence we know of that transport latency is economically material rather than a rounding error, and it is also a demonstration that measuring it requires holding the two feeds side by side.

2.4 The observational setting of this study

Public sportsbook endpoints supply none of that instrumentation. There are no order-level identifiers, because there is no order book to observe; a bookmaker posts a price rather than matching a queue. There is no venue-side timestamp on the price change, only the time at which our own poll retrieved it. There is no second feed from the same book against which dissemination could be differenced, as Ding, Hanna, and Hendershott could difference the consolidated and direct feeds. And the sampling interval is set by the researcher's own polling rather than by the venue's messaging.

The consequence is not that estimation becomes harder. It is that the estimand changes. In the price discovery setting the object of inference is a property of the market, and the apparatus is transparent. Here the apparatus enters the estimand, and the observed quantity is a sum of a market component and two infrastructure components that no amount of additional sampling separates. That is the gap this paper occupies.

2.5 Where this paper sits

The relevant methodological literature is therefore not price discovery but identification. Manski (2003) and Tamer (2010) develop the position we adopt: when a parameter is not point-identified by the available data, the disciplined response is to characterize what the data do support, report bounds where bounds exist, and state the additional information that would restore identification. Applied to transmission, that means treating a latency estimate as a joint statement about a market and about the instrument that observed it, and refusing to report the first without defending the second.

The present paper is motivated not by disagreement with these literatures, but by a different observational setting. Where venue timestamps and order identifiers are available, the questions of Sections 2.2 and 2.3 are answerable as posed and our concerns do not arise. Where they are not, an estimate that would be unremarkable in those settings becomes an inference resting on an unstated assumption about publishing infrastructure. We take the second case seriously because it is the case most outside researchers actually face, and because sportsbook markets make it possible to state the problem precisely rather than in the abstract.

This positioning has a consequence we want to be explicit about. **The contribution stated here does not depend on the sign or significance of any latency estimate we eventually report.** If every estimate proves null, the paper still establishes what a public-endpoint observer can and cannot learn about price formation, which assumptions would have to hold for a leadership claim to be warranted, and what instrumentation a subsequent study would need to build. The identification problem, not the baseball application, is the durable object.

3. Conceptual framework

3.1 The transmission chain

Consider a discrete, publicly observable event in a baseball game occurring at clock time `t_E`: a run scores. The event changes the conditional distribution of the game's final total, and a bookmaker offering a live total on that game will, at some point, revise its posted number. An outside observer records the revision at some later time. Between the event and the record lie three distinct delays, produced by three distinct mechanisms.

Pricing latency, which we write as the interval between the event and the bookmaker's internal decision to revise, is the economically meaningful quantity. It reflects how quickly the bookmaker detects the event, updates its model, and commits to a new number. Differences in pricing latency across books are differences in how the market processes information.

Feed latency is the interval between the bookmaker's internal revision and the moment that revision becomes visible on the public endpoint we query. It is a property of publishing infrastructure, caching, and content delivery, not of judgment about baseball.

Observation latency is the interval between publication and our sampling of it. It is a property of our own collector. With a polling interval of roughly thirty-one seconds, a revision published immediately after a poll waits, on average, about fifteen seconds to be seen, and up to thirty-one seconds in the worst case.

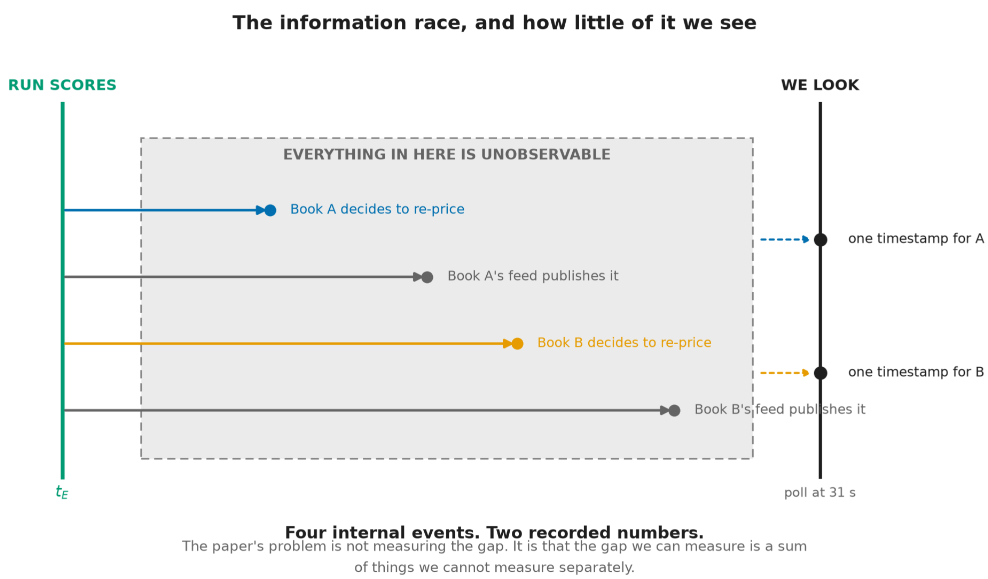

Figure 2. *Four internal events; two recorded numbers.* Each book decides to re-price and each book's feed publishes that decision. None of the four is visible. What we hold is one timestamp per book, and the interval we can measure is

a sum of intervals we cannot measure separately.

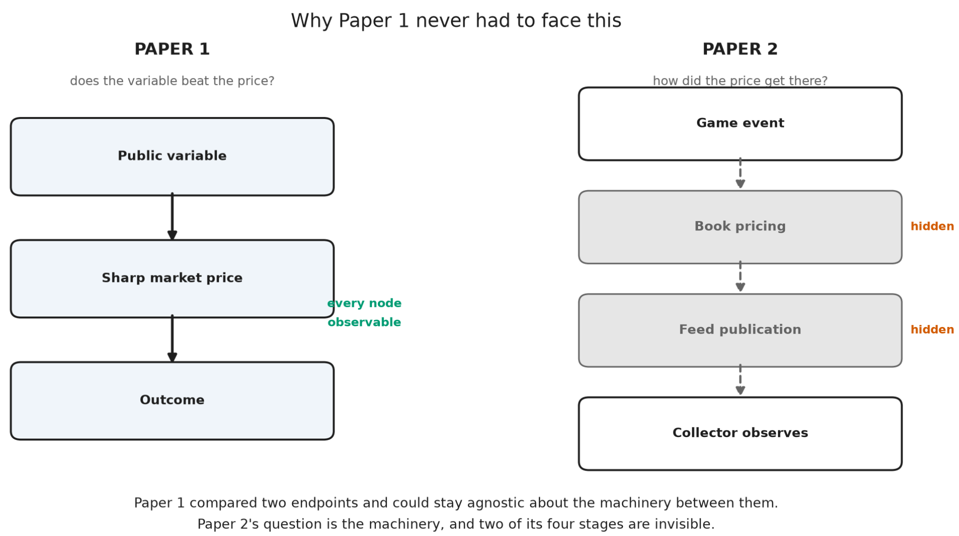

Figure 3. *Paper 1 never needed to identify internal latency.* It compared two endpoints, a public variable and a realized outcome, with the market price between them, and every node it required was observable. This paper's question is the machinery itself, and two of its four stages are hidden from any outside observer.

The data contain only the sum. For book `b` and event `E`, the observable is

```
Δt_b(E) = λ_price_b(E) + λ_feed_b(E) + λ_obs_b(E)
```

As an accounting identity this is unremarkable, and we do not present it as a contribution. Writing a delay as a sum of its parts is bookkeeping. What matters is the consequence: **no manipulation of a single book's series recovers the individual terms, because only the left-hand side is ever observed.** Everything difficult about this paper follows from that sentence, which is why the identification section is longer than the estimation section.

3.2 Why two books might help

If the second and third terms were identical across books, the cross-book difference would remove them:

```
Δt_A(E) - Δt_B(E) = λ_price_A(E) - λ_price_B(E)
```

and the contrast would isolate the economics. This is the hope that motivates a two-book design, and it is an assumption, not a fact.

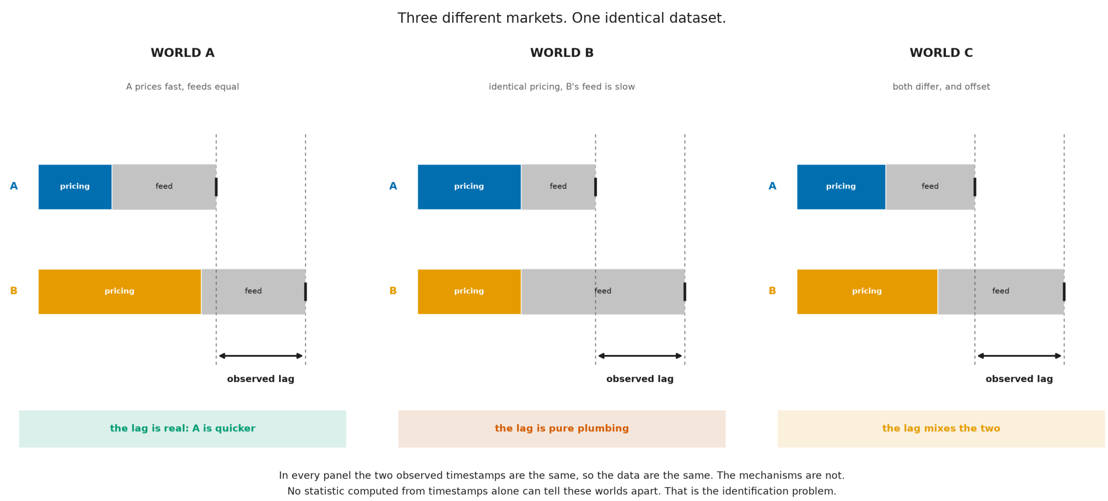

Figure 4. *Identical observations can arise from different underlying markets.* In the first world a bookmaker genuinely prices faster and the publishing infrastructure is equal, so the observed lag means what it appears to mean. In the second, the two bookmakers price at identical speed and the entire observed lag is produced by one book's slower feed. In the third, both differ and partially offset. The observed timestamps, marked by the dashed guides, fall in exactly the same places in all three panels. The datasets are numerically identical; the markets are not. Any statistic computed from timestamps alone assigns the same value to all three, so distinguishing them requires evidence from outside the timing series.

Observation latency is plausibly common-mode by construction: a single collector polls both books on the same schedule, so the sampling penalty is drawn from the same distribution for each. Feed latency is not. Two commercial sportsbooks run different infrastructure, and there is no reason in principle for their publishing delays to match. Whether the difference is material relative to the pricing differences we hope to detect is precisely what Section 3 must establish.

3.3 Why the estimand is anchored to the event

A tempting alternative avoids the event entirely: observe when each book arrives at a given price level and call the earlier one the leader. This is mechanically defective.

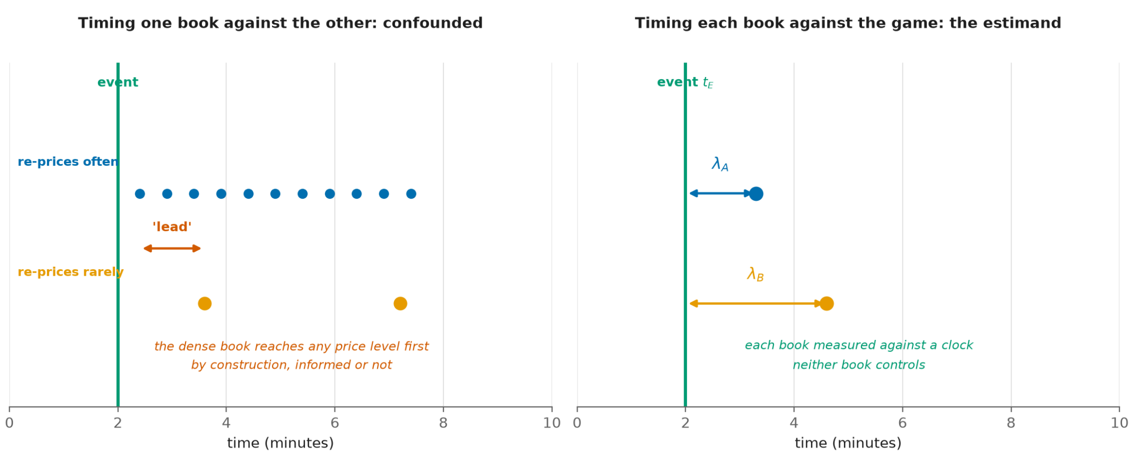

Figure 5. *A denser book leads by construction, not by insight.* Left: a book that re-prices frequently passes through any given price level before a book that re-prices rarely, purely as a consequence of update density. A leadership statistic built on book-to-book arrival times reports this sampling artifact as information leadership. Right: anchoring each book's response to the exogenous game event measures each against a clock that neither book controls, so update density no longer confers a spurious advantage.

The game event is exogenous to both books in the relevant sense: runs are not caused by bookmaker quoting behavior. Anchoring to it converts a comparison between two endogenous, differently sampled series into two comparisons against a common external clock. This does not solve the feed-latency problem, but it removes the frequency confound, which is a distinct and more easily made error.

3.4 What the market is quoting

One structural fact about the instrument shapes the analysis and is easy to get wrong. The posted live number is a **full-game total**, not a forecast of remaining runs: it is the expected final combined score, incorporating runs already scored. Consequently a run that scores enters the posted number with a positive sign, partially offset by the reduction in scoring opportunities that remain. A revision following a run is therefore expected to be upward, and its magnitude is a pass-through fraction rather than an elasticity. We verify this property directly in Section 5 rather than assuming it, because the opposite convention would reverse the sign of every response we measure.

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4. Identification

This section is the paper's core contribution. We state the assumptions under which the cross-book contrast identifies a pricing difference, examine each against what is known about the instrument, and describe what remains estimable when an assumption fails.

4.1 The estimand

As Figure 4 makes clear, the target of inference cannot be read off the data; it must be defined and then argued for. Let `E` index discrete game events with clock times `t_E`, and let `t_b(E)` denote the time of book `b`'s first main-line revision attributable to `E`. Define the event-anchored response latency

```
λ_b = E[ t_b(E) - t_E ]
```

and the cross-book contrast `Δλ = λ_A - λ_B`. The target of inference is `Δλ`, and the question the paper asks is under what conditions `Δλ` is a statement about pricing rather than about plumbing.

4.2 The assumptions

A1. Event exogeneity. Game events are not caused by bookmaker quoting behavior. This is uncontroversial in this setting and is the reason event anchoring is available to us at all.

A2. Clock comparability. Event timestamps and quote timestamps are on a common, comparable clock. This is not automatic. Event times are derived from one upstream provider and quote times from another, and either may be a publication time rather than an occurrence time. A2 requires a direct audit of timestamp provenance, and any residual skew must be quantified rather than assumed away, because a constant skew shifts both `λ_A` and `λ_B` equally but a book-specific skew is indistinguishable from a pricing difference.

A3. A well-defined main line. The response `t_b(E)` presupposes a single price series per book. Books post multiple simultaneous total lines, a main line and alternates, without a discriminator. Reasonable rules for extracting the main line disagree with one another on a large majority of observations, and downstream statistics move materially depending on the choice. A3 therefore requires a fixed, documented, and tested extraction rule, together with a demonstration that the conclusions do not depend on it.

A4. Separability of transport from pricing. This is the binding assumption, and Figure 4 is its statement in visual form. `Δλ` identifies the pricing contrast only if feed latency is common-mode across books, or if a book-specific transport component can be independently measured and removed. We do not assume A4 holds. Establishing whether it holds, or demonstrating rigorously that it cannot be established with instruments of this class, is the paper's principal empirical task.

A5. Sufficiency of two books. With two books, a discrepancy is detectable but not attributable: if one book behaves anomalously there is no third observation to adjudicate which is anomalous. A5 therefore concerns robustness rather than point identification, and it is either satisfied by adding a third live source or addressed by an explicit outlier-detection procedure that does not require one.

A6. Non-arbitrary attribution. The mapping from an event to "the revision caused by it" requires an attribution window. A window chosen after inspecting the data is a researcher degree of freedom capable of manufacturing an effect. A6 is satisfied by pre-registration, which is why the window is fixed in Section 6.5 before any estimate is produced.

4.3 Three admissible outcomes

The design admits exactly three outcomes, and we commit in advance to reporting whichever obtains.

Outcome A. The pricing contrast is identified. The clock audit passes, feed latency is shown to be common-mode or is independently measured, and the extraction rule is shown not to drive the result. The cross-book contrast is then a statement about how two bookmakers process information.

Outcome B. Only bounds are identified. Some assumption holds partially: transport latency is bounded but not known, or the result is directionally stable but magnitude-sensitive to the extraction rule. The reportable object is then an interval within which the pricing contrast must lie, together with the assumption that produced it.

Outcome C. The quantity is not identifiable with this class of instrument. No external measurement of feed latency is obtainable from public endpoints, and no argument establishes common-mode behaviour. This is the case in which the three worlds of Figure 4 remain observationally equivalent no matter how much data accumulates. The reportable result is then a demonstration, not a guess: a proof that timestamp data from public sportsbook endpoints cannot separate market behaviour from publishing infrastructure, together with a specification of the additional instrumentation that would be required.

We want to be explicit that Outcome C is a result and not a failure. Much of the market microstructure literature works in settings where researchers observe richer event and timestamp information than public sportsbook endpoints provide: exchange-side sequencing, order-level identifiers, and venue timestamps that pin the publication stage directly. Where that information is available, the decomposition this paper confronts can be handled by construction. Where it is not, the discipline is to demonstrate its unavailability rather than to proceed as though the richer setting obtained. A paper that establishes where identification breaks down tells a subsequent researcher what to build.

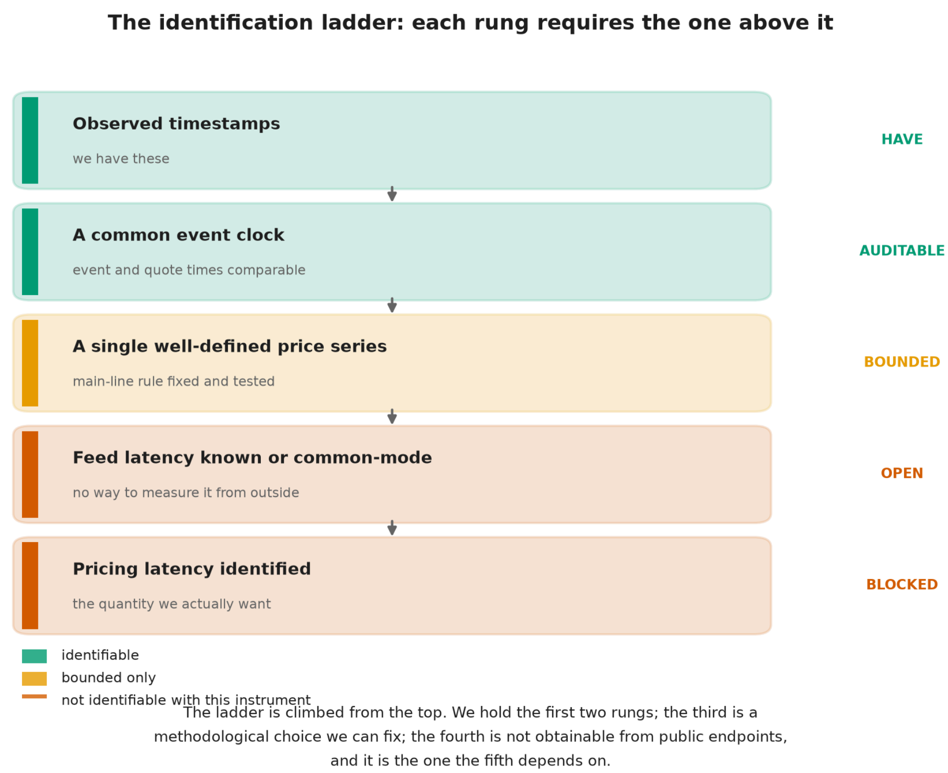

Figure 6. *Identification requires assumptions, not computation.* The requirements form a ladder, and each rung depends on the one above it. Each rung depends on the one above it. We hold the top two. The third is a methodological choice we can fix and test. The fourth, knowledge of feed latency, is not obtainable from public endpoints, and the fifth depends on it. The colour of the fourth rung is what determines whether this paper reports Outcome A, B, or C.

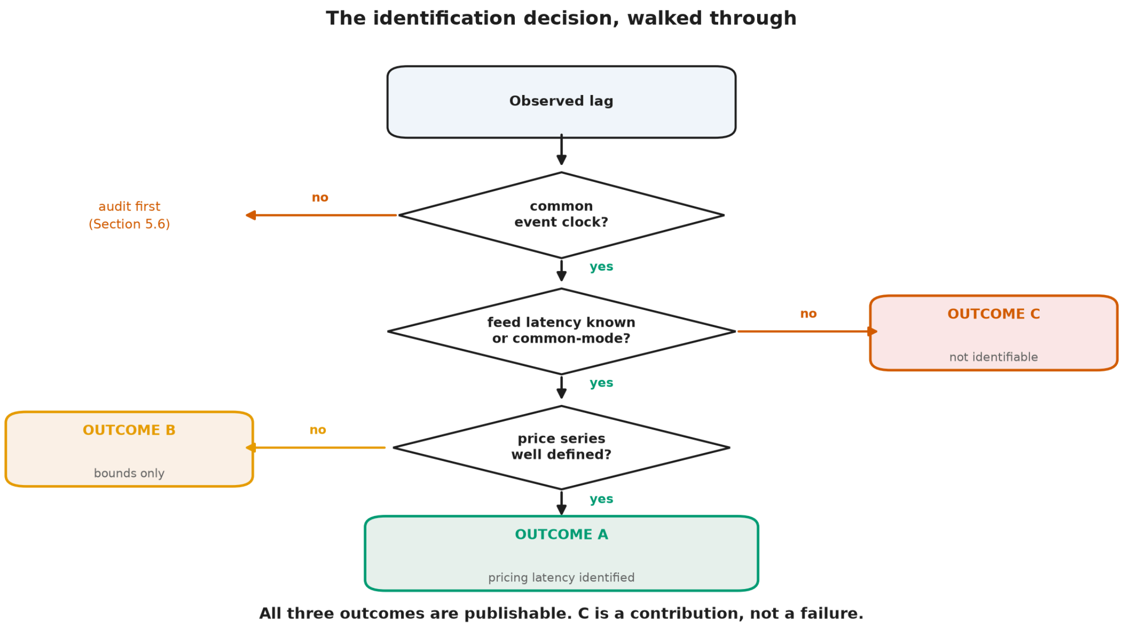

Figure 7. *The paper does not presuppose which branch it takes.* The same logic, as a decision the reader can walk. All three terminal states are publishable, and the third is the one a study is least likely to report.

4.4 The resolution floor

An instrument cannot report structure below its sampling interval.

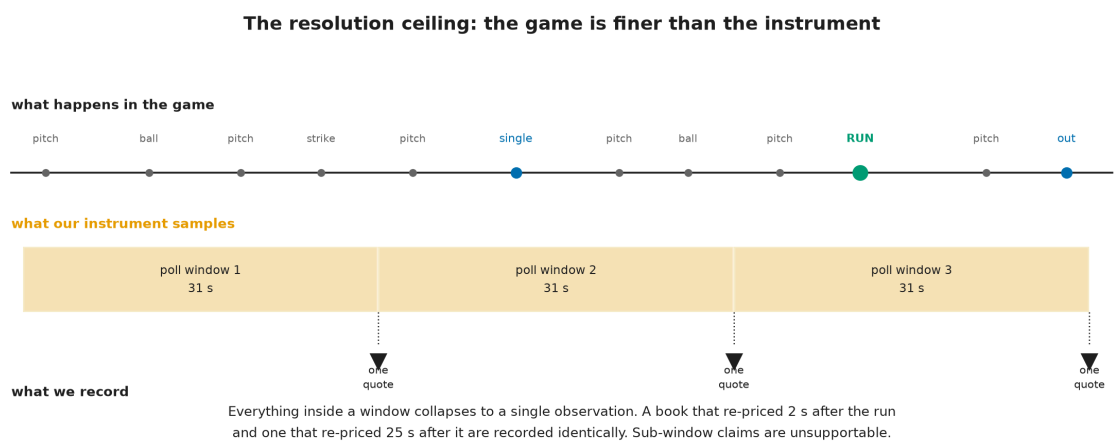

Figure 8. *The instrument itself limits what can be learned.* The game is finer than the sampling window. Pitches, balls, strikes, and the run itself arrive on a scale of seconds; our sampling collapses everything inside a thirty-one-second window into a single observation. A book that re-priced two seconds after the run and one that re-priced twenty-five seconds after it are recorded identically. This bounds the paper's ambition: no claim below the window is supportable, and we make none. The figure describes the apparatus, not the market.

This limit is worth stating plainly because it bounds the paper's ambition. Financial microstructure research often concerns latencies measured in microseconds. Nothing in this design speaks to that regime. What it can speak to is the slower, human-and-model-mediated process by which a sportsbook absorbs a publicly visible event, and whether two such processes can be distinguished from outside.

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5. Data and institutional setting

5.1 The instrument

The data are generated by a continuously running collector that polls two sportsbooks and a game-state provider on a fixed schedule and appends every observation to an append-only panel. The properties below are verifiable facts about the apparatus rather than findings, and we state them as such.

Property	Value
Sampling cadence (live)	approximately 31 seconds, median, both books
Books observed	two, both retail-facing
Games with live quotes	in excess of one hundred, accumulating
Quote fields	posted line, both sides' prices, live flag, market status
Market status coverage	one book only
Game state	inning, half, outs, base occupancy, score, pitch count, times through order
Line semantics	full-game total, verified against pregame distribution

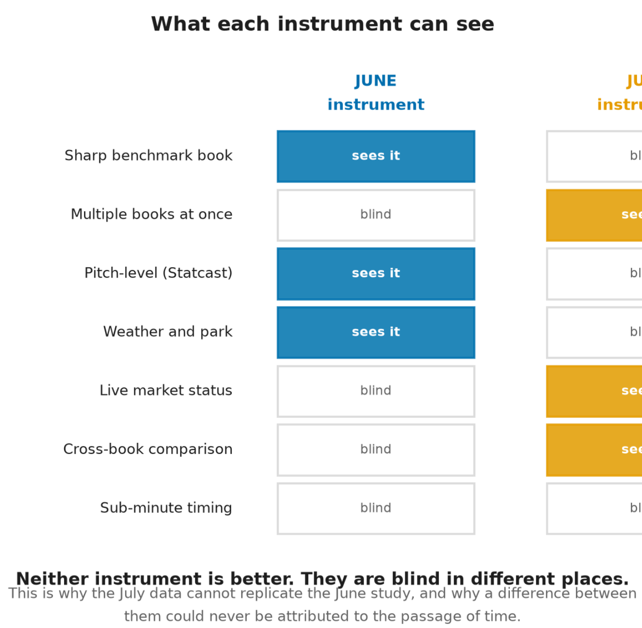

Figure 9. *Different instruments reveal different phenomena.* The June apparatus saw a sharp benchmark book, pitch-level measurement, and weather; it could not see two books at once or a market's live status. The July apparatus sees

the second book and the status stream but is blind to everything the first could measure. Neither is better. They are blind in different places, which is why a result from one cannot confirm or contradict a result from the other.

5.2 What this instrument does not contain

Three absences are load-bearing for interpretation and are properties of the design rather than accidents of a particular sample.

There is **no sharp benchmark book**. Both observed books are retail-facing. The companion study used a sharp book as its incumbent forecast; nothing in this panel plays that role, and the leadership question here is therefore about two retail books rather than about a sharp-to-retail information cascade.

There are **no pitch-level or meteorological covariates**. The companion study's feature set is unavailable here.

Market status is asymmetric. One book publishes an explicit open, suspended, or removed state; the other publishes nothing. Any analysis that filters on tradeability can therefore only be applied to one book, and applying it to one and not the other introduces selection that favors the filtered book.

5.3 Why no comparison with the companion study is possible

It is tempting to treat this dataset as a second sample of the companion study's population. It is not. The two differ in the benchmark book, in the available covariates, and in the sampling cadence. Any difference in results between them would be jointly attributable to the passage of time and to the change of instrument, with no way to apportion between the two. We therefore make no claim in either direction about the companion study's conclusions. A temporal replication of that study requires re-running its own instrument, and is a separate exercise reported elsewhere.

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6. Methods

6.1 Constructing the event series

Events are extracted from the game-state panel as changes in the combined score between consecutive observations. The event time is the timestamp of the first observation exhibiting the new score, which is itself subject to the provider's own publication delay; A2 concerns exactly this, and the audit described in Section 6.6 quantifies it. Events are typed by magnitude, since a solo home run and a bases-clearing double are different information arrivals.

6.2 Constructing the price series

For each book, the main line is extracted per observation timestamp under a single fixed rule, and the series is reduced to the sequence of distinct quote states. Repeated identical quotes are not revisions and are collapsed. The extraction rule is fixed in advance, and the sensitivity of every reported quantity to that choice is reported alongside it, per A3.

6.3 Attribution

A revision is attributed to an event if it is the first distinct main-line change occurring within a fixed post-event window. The window is stated in the pre-registered plan below. Events whose windows overlap a subsequent event are flagged and analyzed separately, since attribution is ambiguous when two information arrivals are closer together than the response time being measured.

6.4 Estimation and inference

The unit of replication is the game, not the observation. Quotes within a game are strongly dependent, and treating them as independent produces intervals that are far too narrow. All estimation therefore proceeds by computing a per-game statistic and treating the game as the sampling unit, with intervals constructed accordingly.

6.5 Pre-registered analysis plan

Every item below exists to prevent one of the failure modes in Figure 4 from being mistaken for a finding. The following is fixed before any estimate is computed. Departures from it will be reported as departures.

1. **Primary estimand.** The cross-book contrast in event-anchored response latency, $\Delta\lambda$.
2. **Event definition.** A change in combined score, typed by magnitude.
3. **Attribution window.** Three hundred seconds after the event. Events with a subsequent event inside the window are analyzed as a separate, flagged stratum.
4. **Main-line extraction.** A single odds-anchored rule, fixed and documented, with a mandatory sensitivity analysis across at least two alternative rules.
5. **Unit of replication.** The game.
6. **Primary comparison.** Direction and magnitude of $\Delta\lambda$, reported with a game-clustered interval.
7. **Mandatory falsification tests.** Before any leadership interpretation is offered: a symmetry test comparing the frequency of pre-event and post-event revisions, which separates a genuine response from a book's baseline revision rate; a shuffled-event placebo; and the extraction sensitivity analysis of item 4.
8. **Reporting rule.** A null result is reported with the same prominence as a positive one. No estimand, window, or extraction rule is added or altered after inspecting the results.

6.6 Conditions required before results may be reported

The Results section of this paper remains unwritten until all four of the following hold, each verified and recorded:

1. **A well-defined main line.** An extraction rule fixed in code and tested, with a demonstration that the primary statistic is materially invariant to reasonable alternatives.
2. **Clock comparability.** An audit establishing that event and quote timestamps are comparable, with residual skew quantified.
3. **Transport separability resolved in one direction.** Either an independent measurement of book-specific feed latency, or a defended argument that it is common-mode, or a documented demonstration that neither is achievable with this class of instrument. The third outcome satisfies this condition and converts the paper into a pure identification result.

4. **Robustness support.** A third live source, or an explicit outlier-detection procedure that does not require one.

Stating these in advance is the point. A paper that fixes its design before seeing its evidence cannot be reshaped by the evidence, and a null that arrives under those conditions carries the same weight as a finding.

7. Scope of the contribution

Here the paper stops being about sportsbooks.

The identification problem set out above is not a peculiarity of betting markets. It arises whenever a researcher observes the output of a process rather than the process itself, and wishes to attribute a measured interval to one stage of that process rather than another. An econometrician timing how quickly an exchange incorporates a macroeconomic release faces the same decomposition: some of the measured delay belongs to the traders, some to the exchange's own dissemination, and some to the vendor feed the researcher happens to have purchased. A political scientist timing how quickly legislatures respond to public opinion confronts it. So does anyone measuring the speed of pass through from a policy rate to a posted lending rate. In each case the estimand is a property of an unobserved stage, and the data are the sum of that stage and the plumbing around it.

What betting markets add is not a novel problem but an unusually clean setting in which to state it. The information arrivals are discrete, publicly visible, and precisely dated; the contracts settle against ground truth within hours; and two competing quotes on the identical contract can be observed simultaneously. Where a general treatment would be forced into abstraction, here the three stages can be named, the boundary of observation can be drawn, and the conditions for identification can be written down and checked. A setting in which the problem is tractable is worth more to the literature than a setting in which it is merely important.

The methodological claim is correspondingly modest and, we think, general. Timestamps are observations; latency is an inference. Any transmission estimate is a joint statement about the market and about the apparatus that observed it, and the two cannot be separated by assertion. Where the separation can be defended, the estimate means what it appears to mean. Where it cannot, the honest report is a bound, or a demonstration that no bound is available, together with a specification of the instrument that would be required. Better measurement produces better identification; better argument does not.

We would rather end on that discipline than on a result. When the mechanism is unobserved, rigour lies not in stronger conclusions but in recognizing precisely where conclusions end.

Observation survives. Identification does not.

Appendix. Where this sits in the research program

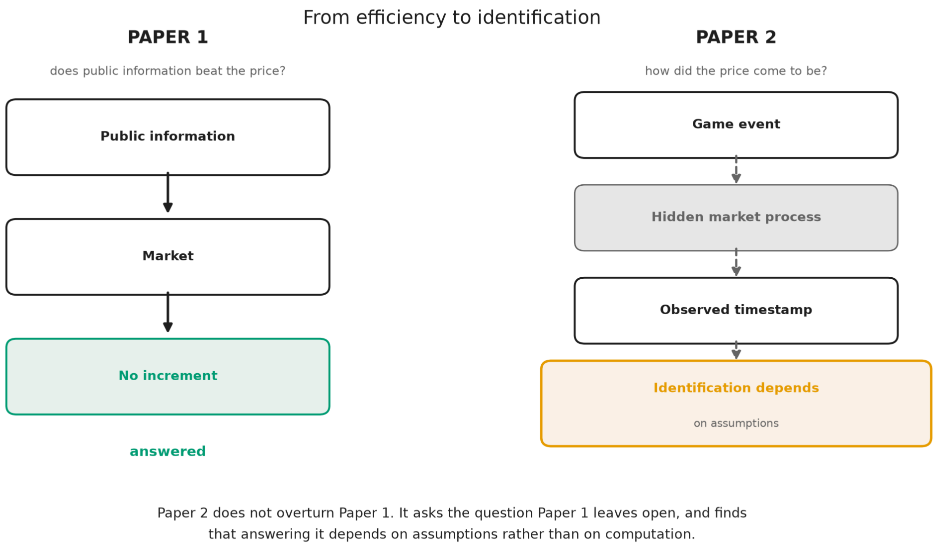

Figure 10. *Information in prices; formation of prices.* Paper 1 asked whether prices contain public information and could answer it, because every node it needed was observable. This study starts from an event, passes through a process it cannot see, and arrives at a timestamp whose interpretation depends on assumptions rather than on computation. The first paper ends in a finding. The second ends, at best, in a set of conditions.

Sections 8 (Results) and 9 (Discussion) are intentionally not drafted. See Section 6.6.

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